

 RV UNIVERSITY <i>Go, change the world</i> an initiative of RV EDUCATIONAL INSTITUTIONS	USN:									
	School of Computer Science and Engineering B.Tech (Hons.) CIE-3 Academic Year 2024 – 2025									
	Course: Calculus and Laplace Transform			Course Code: CS2802			Semester: IV			
	Date: 11 April 2025			Duration: 60 Minutes			Max Marks: 15 Marks			

Instructions:

- All questions are compulsory.
- **Write your USN in the space provided** as soon as you get the question paper.
- Do not answer any part of the answers in the question paper.
- Do not engage in any unfair practices.
- Non-programmable calculators are allowed.

S.No	Question	Marks	Level	CO
1	a) Calculate the gradient of the function $f(x, y) = x^3 + 4xy^2 - x$. [2] b) Starting from an initial guess of $(0, 0.25)$, evaluate the first two steps of the gradient descent method for the function. Take the step-size (learning rate) to be 0.1 [3]	5	L3	CO 3
2	a) Let $f(x, y) = e^x \sin(y)$. Find the second-order Taylor series expansion of $f(x, y)$ about the point $(0, 0)$. [3] b) Use the expansion to approximate the value of $f(0.1, 0.2)$ up to second-order terms and also determine the tangent plane at the point. [2]	5	L3	CO 3
Continued on next page...				

3	a) Evaluate the Laplace transform of $f(t) = e^{2t} \cosh 2t$ where $\cosh(2t)$ is the hyperbolic cosine function. [2] b) Using the properties of linearity of Laplace Transforms calculate the Laplace transform of the function: . [3] $f(t) = 2t^2 - \sin(t) + 4e^{-2t}; \quad 0 \leq t < \infty$	5	L3	CO 4
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Course Outcomes	
CO1:	Understand foundational concepts of functions, limits, continuity and derivatives to solve various engineering problems.
CO2:	Apply derivatives and integrations in problems including rate of change, surface areas, volumes, probabilities, and single variable optimization.
CO3:	Solve linear approximations and unconstrained optimization problems related to engineering applications.
CO4:	Apply Laplace transform to solve problems on time varying signals.

Marks Distribution									
L1	L2	L3	L4	L5	L6	CO1	CO2	CO3	CO4
-	-	15	-	-	-	-	-	10	5

End

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Answer Keys:

Q.1)

- a) Calculate the gradient of the function $f(x, y) = x^3 + 4xy^2 - x$. [2]

Solution:

$$f(x, y) = x^3 + 4xy^2 - x$$

$$\Rightarrow \nabla f = \begin{bmatrix} 3x^2 + 4y^2 - x \\ 8xy \end{bmatrix}$$

- b) Starting from an initial guess of $(0, 0.25)$, evaluate the first two steps of the gradient descent method for the function. Take the step-size (learning rate) to be 0.1 [3]

Solution: Given the initial guess point, $P_0 = (0, 0.25) = (0, 1/4)$, the gradient descent technique is given by:

$$P_{n+1} = P_n - \alpha \nabla f(P_n)$$

$$P_0 = \begin{bmatrix} 0 \\ 1/4 \end{bmatrix}$$

$$\nabla f(P_0) = \begin{bmatrix} \frac{4}{4^2} - 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{-3}{4} \\ 0 \end{bmatrix}$$

$$\therefore P_1 = P_0 - 0.1 \nabla f(P_0)$$

$$P_1 = \begin{bmatrix} 0 \\ \frac{1}{4} \end{bmatrix} - \frac{1}{10} \begin{bmatrix} \frac{-3}{4} \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{3}{40} \\ \frac{1}{4} \end{bmatrix}$$

Now,

$$P_2 = P_1 - \frac{1}{10} \nabla f\left(\frac{3}{40}, \frac{1}{4}\right)$$

$$P_2 = \begin{bmatrix} \frac{3}{40} \\ \frac{1}{4} \end{bmatrix} - \frac{1}{10} \begin{bmatrix} -0.73 \\ 0.15 \end{bmatrix} = \begin{bmatrix} 0.148 \\ 0.235 \end{bmatrix}$$

Q2).:

- a) Let $f(x, y) = e^x \sin(y)$. Find the **second-order Taylor series expansion** of $f(x, y)$ about the point $(0, 0)$. [3]

Solution:

Taylor formula for $f(x, y)$ about a point (x_0, y_0) up to 2^{nd} order is:

$$f(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + \frac{1}{2} (f_{xx}(x_0, y_0)(x - x_0)^2 + 2f_{xy}(x_0, y_0)(x - x_0)(y - y_0) + f_{yy}(x_0, y_0)(y - y_0)^2)$$

We have:

$$\begin{aligned} f(0, 0) &= 0 \\ f_x &= e^x \sin(y); \Rightarrow f_x(0, 0) = 0 \\ f_{xx} &= e^x \sin y; \Rightarrow f_{xx}(0, 0) = 0 \\ f_y &= e^x \cos(y); \Rightarrow f_y(0, 0) = 1 \\ f_{yy} &= -e^x \sin y \Rightarrow f_{yy}(0, 0) = 0 \\ f_{xy} &= e^x \cos y \Rightarrow f_{xy}(0, 0) = 1 \end{aligned}$$

Thus, the Taylor formula (Maclaurin) is:

$$\mathbf{f}(\mathbf{x}, \mathbf{y}) = \mathbf{y} + \mathbf{xy}$$

- b) Use the expansion to approximate the value of $f(0.1, 0.2)$ up to second-order terms and also determine the tangent plane at the point. [2]

Solution:

From the above, the quadratic approximation of the surface at $(0, 0)$ is given by:

$$f(x, y) = y + xy$$

.

Using this, the approximate value of the function at $(0.1, 0.2)$ is:

$$f(0.1, 0.2) = 0.2 + 0.1 \times 0.2 = \mathbf{0.22}$$

From the original function:

$$f(0.1, 0.2) = e^{0.1} \sin 0.2 = \mathbf{0.2196}$$

Knowing that the quadratic approximation is $f(x, y) = y + xy$, the Tangent plane is given by the Taylor formula up to the linear term. Hence:

$$\mathbf{T}(\mathbf{x}, \mathbf{y}) = \mathbf{y}$$

Q 3).

- a) Evaluate the Laplace transform of $f(t) = e^{2t} \cosh 2t$ where $\cosh(2t)$ is the hyperbolic cosine function.
[2]

Answer: We know:

$$\mathcal{L}\{\cosh(2t)\} = \frac{s}{s^2 - 4}$$

Using the shift property:

$$\mathcal{L}\{e^{2t} \cosh(2t)\} = \frac{s}{s^2 - 4} \Big|_{s \rightarrow s-2} = \frac{s-2}{(s-2)^2 - 4}, \quad s > 4$$

- b) Using the properties of linearity of Laplace Transforms calculate the Laplace transform of the function: . [3]

$$f(t) = 2t^2 - \sin(t) + 4e^{-2t}; \quad 0 \leq t < \infty$$

Answer: By Applying the Linearity property

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{2t^2\} - \mathcal{L}\{\sin(t)\} + \mathcal{L}\{4e^{-2t}\}$$

$$\mathcal{L}\{f(t)\} = 2\mathcal{L}\{t^2\} - \mathcal{L}\{\sin(t)\} + 4\mathcal{L}\{e^{-2t}\}$$

and then using the Standard Laplace Transforms

$$\begin{aligned} \mathcal{L}\{t^2\} &= \frac{2}{s^3} \\ \mathcal{L}\{\sin(t)\} &= \frac{1}{s^2+1} \\ \mathcal{L}\{e^{-2t}\} &= \frac{1}{s+2} \end{aligned}$$

Substituting and then simplifying this

$$\begin{aligned} \mathcal{L}\{f(t)\} &= 2 \cdot \frac{2}{s^3} - \frac{1}{s^2+1} + 4 \cdot \frac{1}{s+2} \\ &= \frac{4}{s^3} - \frac{1}{s^2+1} + \frac{4}{s+2} \end{aligned}$$

End

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S.No	Question	Marks	Level	CO
1	Using the gradient descent technique, minimize the function $f(x, y) = (x - 2)^2 + (y - 3)^2$ starting from the initial guess $(x_0, y_0) = (0, 0)$ and using a learning rate $\alpha = 0.5$. Show that the method converges to the minimum in two steps.	5	L3	CO 3
2	a) Evaluate the Taylor series of the function $g(x, y) = \ln(x + y)$ about the point $P_0 = (1, 1)$ up to second order in x and y. [3] b) Using the Taylor formula calculated above, deduce the tangent plane of the function about $(1, 1)$ and obtain the approximate value of $g(1.1, 1.1)$ using it. [2]	5	L3	CO 3
Continued on next page...				

3	a) Evaluate the Laplace transform of $f(t) = e^{2t} \sinh 2t$ where $\sinh(2t)$ is the hyperbolic sine function. [2] b) Using the properties of linearity of Laplace Transforms calculate the Laplace transform of the function: . [3] $f(t) = 4t^2 - 3 \cos(t) + 5e^{-t}$	5	L3	CO 4
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Answer Keys:

Q 1). Using the gradient descent technique, minimize the function

$$f(x, y) = (x - 2)^2 + (y - 3)^2$$

starting from the initial guess $(x_0, y_0) = (0, 0)$ and using a learning rate $\alpha = 0.5$. Show that the method converges to the minimum in two steps.

Solution:

$$f(x, y) = (x - 2)^2 + (y - 3)^2$$

$$\Rightarrow \nabla f = \begin{bmatrix} 2(x - 2) \\ 2(y - 3) \end{bmatrix}$$

Given the initial guess point, $P_0 = (0, 0)$, the gradient descent technique is given by:

$$P_{n+1} = P_n - \alpha \nabla f(P_n)$$

$$P_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\nabla f(P_0) = \begin{bmatrix} -4 \\ -6 \end{bmatrix}$$

$$\therefore P_1 = P_0 - 0.5 \nabla f(P_0)$$

$$P_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -4 \\ -6 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$P_2 = P_1 - \frac{1}{2} \nabla f(2, 3)$$

$$P_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

We observe that $P_2 = P_1$, which means that the iteration is no longer changing and convergence is reached and $(2, 3)$ is the minima. This can also be easily seen from the first derivative test that $f(x, y)$ has minima at $(2, 3)$

Q 2).

- a) Evaluate the Taylor series of the function $g(x, y) = \ln(x + y)$ about the point $P_0 = (1, 1)$ up to second order in x and y . [3]

Solution:

Taylor series of $g(x, y)$ up to second order about (x_0, y_0) is given by:

$$g(x, y) = g(x_0, y_0) + g_x(x_0, y_0)(x - x_0) + g_y(x_0, y_0)(y - y_0) + \frac{1}{2} (g_{xx}(x_0, y_0)(x - x_0)^2 + 2g_{xy}(x_0, y_0)(x - x_0)(y - y_0) + g_{yy}(x_0, y_0)(y - y_0)^2)$$

For $g(x, y) = \ln(x + y)$, and $(x_0, y_0) = (1, 1)$, we have,

$$\begin{aligned} g(1, 1) &= \ln 2 \approx 0.6931 \\ g_x &= \frac{1}{x + y} \Rightarrow g_x(1, 1) = \frac{1}{2} \\ g_y &= \frac{1}{x + y} \Rightarrow g_y(1, 1) = \frac{1}{2} \\ g_{xx} &= \frac{-1}{(x + y)^2} \Rightarrow g_{xx}(1, 1) = \frac{-1}{4} \\ g_{yy} &= \frac{-1}{(x + y)^2} \Rightarrow g_{yy}(1, 1) = \frac{-1}{4} \\ g_{xy} &= \frac{-1}{(x + y)^2} \Rightarrow g_{xy}(1, 1) = \frac{-1}{4} \end{aligned}$$

Therefore, the Taylor Formula for $g(x, y) = \ln(x + y)$ about $(1, 1)$ up to quadratic term is:

$$\begin{aligned} g(x, y) &= \ln 2 + \frac{1}{2}(x - 1) + \frac{1}{2}(y - 1) + \frac{1}{2} \left(\frac{-1}{4}(x - 1)^2 + 2 \times \frac{-1}{4}(x - 1)(y - 1) - \frac{1}{4}(y - 1)^2 \right) \\ \Rightarrow g(x, y) &= x + y - \frac{1}{8} \left(x^2 + y^2 - \frac{3}{2} + \ln 2 \right) \end{aligned}$$

- b) Using the Taylor formula calculated above, deduce the tangent plane of the function about $(1, 1)$ and obtain the approximate value of $g(1.1, 1.1)$ using it. [2] From the Taylor formula in (a), the Tangent plane about $(1, 1)$ is given by:

$$T(x, y) = \ln 2 + \frac{1}{2}(x - 1) + \frac{1}{2}(y - 1)$$

The linear approximation at $(1.1, 1.1)$ is:

$$\mathbf{T(1.1, 1.1) \approx 0.7931}$$

.

The actual value of the function at $(1.1, 1.1)$ is $g(1.1, 1.1) = \ln(1.1 + 1.1) = \ln 2.2 \approx \mathbf{0.7885}$.

Q3).

- a) Evaluate the Laplace transform of $f(t) = e^{2t} \sinh 2t$ where $\sinh(x)$ is the hyperbolic sine function. [2]

Answer: We know:

$$\mathcal{L}\{\sinh(2t)\} = \frac{2}{s^2 - 4}$$

Using the shift property:

$$\mathcal{L}\{e^{2t} \sinh(2t)\} = \frac{2}{s^2 - 4} \Big|_{s \rightarrow s-2} = \frac{2}{(s-2)^2 - 4}, \quad s > 4$$

- b) Using the properties of linearity of Laplace Transforms calculate the Laplace transform of the function: . [3]

$$f(t) = 4t^2 - 3 \cos(t) + 5e^{-t}$$

Answer: By Applying the Linearity property

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{4t^2\} - \mathcal{L}\{3 \cos(t)\} + \mathcal{L}\{5e^{-t}\}$$

$$\mathcal{L}\{f(t)\} = 4\mathcal{L}\{t^2\} - 3\mathcal{L}\{\cos(t)\} + 5\mathcal{L}\{e^{-t}\}$$

and then using the Standard Laplace Transforms

$$\begin{aligned} \mathcal{L}\{t^2\} &= \frac{2}{s^3} \\ \mathcal{L}\{\cos(t)\} &= \frac{s}{s^2+1} \\ \mathcal{L}\{e^{-t}\} &= \frac{1}{s+1} \end{aligned}$$

Substituting and then simplifying this

$$\begin{aligned} \mathcal{L}\{f(t)\} &= 4 \cdot \frac{2}{s^3} - 3 \frac{s}{s^2+1} + 5 \cdot \frac{1}{s+1} \\ &= \frac{8}{s^3} - \frac{3s}{s^2+1} + \frac{5}{s+1} \end{aligned}$$

End
